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Evaluating provider report of fidelity to contingency management in opioid treatment programs

Elizabeth Casline[a,*], **Kelli Scott**[a], **Cara M. Murphy**[b], **Bryan R. Garner**[c], **Sara J. Becker**[a]

[a]Institute for Public Health and Medicine, Center for Dissemination and Implementation Science, Northwestern University Feinberg School of Medicine, 633 N. St Clair, Chicago, IL 60611, USA

[b]Department of Behavioral & Social Science, Brown University School of Public Health, Box G-S121-5, Providence, RI 02912, USA

[c]Division of General Internal Medicine, Department of Internal Medicine, The Ohio State University College of Medicine, 2050 Kenny Road, Columbus 43221, USA

Abstract

Introduction: With growing adoption of contingency management (CM) in addiction treatment programs, ensuring intervention fidelity over time is essential for improving patient outcomes. Nonetheless, ensuring an intervention is delivered as intended can be time- and resource-intensive for organizations. Finding ways to monitor fidelity without unduly burdening health systems is critical.

Methods: This study evaluated the feasibility of using provider report to monitor CM fidelity compared to traditional observer ratings using the CM Competence Scale, leveraging data from 28 opioid treatment programs that participated in a hybrid implementation-effectiveness trial. Providers (n = 86) reported CM fidelity across 3143 sessions with observer ratings conducted for 72 of these sessions from 29 providers to assess concurrence of provider- and observer-ratings.

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[*]Correspondence to: 633 N. St Clair, Suite 2000, Chicago, IL 60611, USA. elizabeth.casline@northwestern.edu (E. Casline).

Contributors

EC generated the hypotheses, led the analysis and created a first draft of the introduction, results, and tables: she was also responsible for compiling the sections and ensuring a cohesive voice. SB, KS, and CM drafted the introduction, discussion, and abstract, respectively. SB and BG conceptualized and acquired funding for Project MIMIC and KS and CM were Co-Investigators responsible for coding contingency management recordings (CM was the consensus coder). All authors contributed to drafting and revising the article and have approved the final manuscript.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sara Becker and Bryan Garner reports financial support was provided by National Institute on Drug Abuse. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

CRediT authorship contribution statement

Sara J Becker: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Bryan R Garner:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Cara M Murphy:** Writing – review & editing, Writing – original draft, Methodology. **Kelli Scott:** Writing – review & editing, Writing – original draft, Methodology. **Elizabeth Casline:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.drugalcdep.2024.112544.

Results: Providers reported high fidelity for most CM practices, with high concordance with ratings from trained observers on practices that were easily observable/objective (e.g., discussing reinforcement earned in current and future session). In contrast, concordance between provider and observer ratings was lower for more nuanced practices (e.g., making connections between CM and the patients' broader treatment and recovery goals).

Conclusions: Overall, our findings suggest that while provider-report may effectively capture many aspects of CM delivery, discrepancies in fidelity reporting of specific CM practices warrant further investigation. Future research is needed to determine the optimal approaches for ensuring providers consistently deliver all CM elements with fidelity.

Keywords

Contingency management; Opioid use disorder; Intervention fidelity

1. Introduction

Contingency Management (CM) is an evidence-based behavioral intervention for substance use disorders that uses tangible rewards to reinforce patients for meeting treatment goals (Bentzley et al., 2021; Sheridan Rains et al., 2020). CM is a highly effective adjunctive intervention to medication for opioid use disorder (Bolívar et al., 2021), yet is underutilized in opioid treatment programs (OTPs; Park et al., 2024). Efforts to address this research-to-practice gap have recently been guided by implementation science which uses multi-level strategies to address provider- and organization-level barriers to CM adoption (Becker et al., 2023). An overarching goal of these implementation strategies is to ensure providers deliver CM with fidelity, meaning CM practice components are consistently delivered as intended (Carroll, 2000). To reach this goal, a training-to-criterion approach is frequently used which requires providers to meet a skills-based fidelity criterion prior to CM delivery. Provider practices are then monitored and feedback about provider drift away from criterion levels is given. This skills-based approach has been found to significantly improve provider competency in CM delivery throughout implementation (Hartzler et al., 2014; Petry et al., 2012a).

Widespread use of this training-to-criterion approach is hindered by the reliance on observational coding systems to evaluate provider fidelity. The CM Competence Scale (CMCS) is a gold-standard observational rating scale to evaluate CM fidelity, with strong conceptual and psychometric support (Petry et al., 2010; Petry and Ledgerwood, 2010), yet observational systems are time- and resource-intensive for organizations to complete (DePhilippis et al., 2018). Pragmatic, psychometrically valid measures of fidelity are needed to support CM implementation in real world, lower-resourced settings (Schoenwald et al., 2011). Prior pragmatic approaches to monitoring CM fidelity have included monthly organizational reports (DePhilippis et al., 2018) and written CM vignettes to assess communication skills (Hartzler, 2015). These approaches captured meaningful variability in CM fidelity but did not allow for session-by-session evaluation, a key component for evaluating training and treatment effectiveness.

Provider-report of CM session fidelity has the potential to balance rigor and practicality but has not yet been evaluated. In general, provider-report tends to overestimate adherence to therapy practices (McLeod et al., 2023), however, some studies suggest reporting on more objective practices like those used in CM (e.g., administering reinforcement/rewards) can improve therapist reporting accuracy (Hogue et al., 2015). This secondary analysis assessed the utility of provider-report of CM fidelity using data from Project MIMIC (Maximizing Implementation of Motivational Incentives in Clinics), a cluster-randomized type 3 hybrid implementation-effectiveness randomized controlled trial to improve CM implementation in OTPs (clinicaltrials.gov NCT03931174; Becker et al., 2021). We evaluated whether a brief, provider-report tool could be used as a reasonable alternative to time-intensive observer ratings of CM fidelity, by examining the concurrence of provider-report and observer-rated sessions.

2. Materials and methods

2.1. Parent study

Project MIMIC included 28 OTPs, which contained 186 OTP providers with active caseloads, that were cluster-randomized to receive one of two multi-level implementation strategies: the Addiction Technology Transfer Center (ATTC; workshop training + performance feedback + coaching) or the Enhanced ATTC (E-ATTC; ATTC strategy + Pay for Performance Incentives + Implementation and Sustainment Facilitation). Project MIMIC implemented prize-based CM with escalating prize draws targeting treatment attendance to any aspect of OTP treatment (e.g., medication dosing, individual counseling, or group counseling appointments; Becker et al., 2021). All providers gave written informed consent, in concordance institutional review board-approved procedures (Brown University #1811002260)

2.2. CM fidelity criterion training

Providers in both conditions attended 5–6 h of workshop training consisting of didactic instruction in CM, small group discussion, trainer CM demonstration/modeling, and behavioral rehearsal, all of which emphasized CM delivery to a pre-established fidelity criterion (see Hartzler, 2023). This criterion was based on the CMCS (Petry and Ledgerwood, 2010), a 9-item observer report that assesses six core CM-specific practices and three general clinical practices on a scale from 1 (not at all competent) to 7 (highest degree of competence) (Petry et al., 2010). The six core CM practices were: informing the patient of reinforcement earned in the current session; informing the patient of reinforcement possible in the next session; administering reinforcement; complimenting or praising the patient’s efforts towards attendance; assessing the patient’s desire for prizes; and linking attendance and the CM program to abstinence and other recovery goals (Supplemental Table 1). Providers were trained to recognize when delivery of each CM practice was “adequate” or higher (i.e., CMCS score ≥ 4.0) (Petry and Ledgerwood, 2010). After the workshop, providers submitted an audio recorded CM role play. Providers were required to meet the fidelity criterion of an average CMCS score ≥ 4.0 across all nine items to be approved to deliver CM. Providers whose first role play did not meet this criterion were provided personalized feedback and resubmitted role plays until the criterion was met.

2.3. CM fidelity data collection

2.3.1. Provider-rated fidelity—Each provider tracked their CM fidelity while providing CM to patients over a 9-month period. After each session, providers completed a brief online checklist indicating whether they delivered each of the six CM-specific practices using dichotomous "Yes/No" questions (Supplemental Figure 1).

2.3.2. Observer-rated fidelity—On a monthly basis, providers were encouraged to submit an audio recording of at least one session for performance feedback. Providers in the E-ATTC condition earned $50 if the session had an average CMCS rating of 5.8, which was the average fidelity attained by highly trained counselors in prior effectiveness trials (Petry et al., 2012a). Each submitted recording was coded for fidelity by independent Bachelors-level research staff, masked to condition, and trained to rate audio recordings using the CMCS. Coder training included viewing the CM workshop training, reading the CMCS coding manual, and consensus coding at least two recordings with a PhD-level researcher with CM expertise until agreement (defined as within plus or minus 1 on the 7-point Likert scale) for each CMCS item was obtained. Over 50 % of recordings were double coded with inter-rater agreement of 89 %.

2.4. Analysis plan

2.4.1. Preliminary analyses—Descriptive statistics of CMCS items (i.e., mean, standard deviation, range) were generated for provider- and observer-reported ratings. We assessed whether provider-reported ratings systematically differed as a function of whether providers submitted recordings, implementation strategy condition (ATTC vs. E-ATTC) and a range of provider characteristics (e.g., age, level of experience, gender, race/ethnicity).

2.4.2. Analysis of primary aims—To assess concordance of provider- and observer-report, observer-rated CMCS scores were dichotomized as "Criterion Met" (item rating ≥ 4.0) or "Not Met" (item rating < 4.0). Percent agreement between provider and observer ratings was calculated for each item, kappa statistics were not used due to high rates of provider-reported CM practice delivery.

3. Results

Of the 186 enrolled providers, 86 (46.2 %) delivered CM to at least one patient, and 100 % completed at least one provider-report of fidelity (completed for 3143 sessions from 444 unique patients). Only 29 (33.7 %) providers submitted an audio recorded session for observational coding (74 sessions from 56 unique patients). Of the 74 observer-coded sessions, 72 (97.3 %) had a corresponding provider-report (28 unique providers and 55 unique patients): these 72 sessions comprise the primary analytic sample. Comparison of providers who did and did not submit both provider-report and an audio recording found no differences as a function of condition or provider characteristics (Table 1); hence, comparison of provider- and observer-report pools data across 72 sessions.

3.1. CM practice delivery

Table 2 depicts the number of provider- and observer-rated sessions that met the fidelity criterion for each CMCS item and for the overall scale. Across the 3143 sessions with provider-report, providers reported adequately delivering each CM practice nearly all the time (range: 97–100 %) such that 96 % of sessions reportedly covered all six practices. Virtually identical rates were reported in the 72 sessions with audio recordings (range: 97–100 %, 96 % covering all six practices).

Across the 72 observer-rated sessions, four of the CM practices (items 1 – 3, 5) had high rates of “adequate” delivery (range: 90–99 %), whereas two of the CM practices (items 4 & 6) were rated “adequate” less often (85 % and 58 %, respectively). Combined, only 49 % of sessions demonstrated “adequate” delivery of all six CM practices according to observer report. CMCS overall average session scores ranged from 3.22 – 6.56 (Supplemental Table 1); notably, only one session failed to meet the fidelity criterion.

3.2. Provider-observer agreement

Agreement between provider- and observer-rated practices ranged from high (90–99 %; items 1 – 3, 5), to moderate (83 %; item 4), to poor (62 %; item 6). Agreement was highest (97–99 %) for the three practices focused on reinforcement (items 1 – 3; Table 2). The lowest agreement between provider- and observer-report was for the practice tying attendance to treatment goals (item 6; Table 2). In all but one instance, discordance was due to the observer indicating a CM-specific practice was below the level needed for adequate fidelity when the provider reported the practice was demonstrated. In a fair number of discordant cases, the observer rated provider delivery as “3: somewhat consistent… clearly not adequate” suggesting the provider did address the intervention element but did so in a way only somewhat consistent with the CM approach (Supplemental Table 1).

4. Discussion

Comparison of provider- and observer-reported fidelity revealed that providers were able to report four CM practices with a high degree of accuracy: informing patients of reinforcement earned in the session, highlighting reinforcement that could be earned in the next session, administering reinforcement, and praising patient efforts to attend treatment. Providers reported on assessing desire for prizes with moderate accuracy. In contrast, provider reports of linking attendance to recovery goals were not well aligned with observer report. These findings mirror the broader literature which suggests providers tend to overreport practice use but may be more accurate when reporting on practices that are planned, structured, and easily observable (Hogue et al., 2015; McLeod et al., 2023). Notably, most of the recorded sessions met the fidelity criterion overall, indicating that low fidelity to a specific CM-practice did not reflect low fidelity to CM overall. These results suggest the training enabled providers to accurately report on most CM practices, indicating more intensive, observer rated fidelity may not be required for those practices. Future work should evaluate whether training on linking attendance with recovery would improve provider-reported fidelity.

A key limitation was reliance on audio recordings from a self-selected sample of providers, as providers who submitted recordings may have been more skilled than providers who did not submit. Providers may also have submitted CM recordings they felt were particularly high quality, therefore CM fidelity may have been lower in sessions not captured in this analysis. Moreover, the use of a dichotomous provider rating (yes/no) limits the sensitivity of provider-report and confounds comparison to observer ratings, which allowed for one point variability in coder reliability. Finally, our only two sources of CM session data were provider-report and recordings, which limits our ability to confirm whether all CM sessions were captured. Future work should seek to confirm CM delivery via medical record data to bolster confidence in the delivery rates reported here.

5. Conclusion

The inconsistent concurrence between provider- and observer-report revealed that certain CM reinforcement-related practices may be particularly well-suited for provider-report, whereas other CM practices may be vulnerable to overreporting of fidelity. More work is needed to determine the optimal approach to training providers to CM fidelity criterion, for monitoring fidelity over time, and for ensuring CM implementation results in improved clinical outcomes. The extant literature provides evidence that the total CMCS score is predictive of longer duration of abstinence (Petry et al., 2010, 2012b), and future research should explore whether each CM practice individually predicts patient outcomes to further optimize the CMCS criterion score. Findings from such research may then inform which CM practices are most important and whether provider-reported fidelity is sufficient for clinical and research use. If CM practices that had lower observer-rated CM fidelity (e.g., linkage of CM attendance to abstinence and other recovery goals) are associated with patient outcomes, providers may require more extensive training to consistently deliver these elements with fidelity, including ongoing supervision with active fidelity monitoring (Petry et al., 2012b). Overall, findings highlight future directions to inform CM fidelity assessment as it is being actively scaled to enhance OTP care and improve patient outcomes.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Table 1

Demographic characteristics of providers who submitted at least one contingency management fidelity rating.

	Overall (N = 86)	Submitted At Least One Audio Recorded Session		Statistical Comparison[1]
		Yes (n = 28)	No (n = 58)	
Demographics				
Age	39.50 (12.06, 22 – 71)	36.82 (11.11, 22–65)	40.79 (13.81, 23–71)	$t(84) = 1.33$, $p = 0.19$
White, non-Hispanic	63 (73.3 %)	23 (82.1 %)	40 (69.0 %)	$\chi^2(1) = 1.67$, $p = 0.30$
Cisgender Female	69 (80.2 %)	23 (82.1 %)	46 (79.3 %)	$\chi^2(1) = 0.10$, $p = 1.00$
Clinical Characteristics				
Master's Degree or Higher	51 (59.3 %)	20 (71.4 %)	31 (53.4 %)	$\chi^2(1) = 2.53$, $p = 0.16$
Licensed or Certified Addiction Counselor	45 (52.3 %)	19 (67.9 %)	26 (44.8 %)	$\chi^2(1) = 4.02$, $p = 0.07$
Years Experience	8.33 (9.02, 0–40)	7.52 (7.3, 1–31)	8.72 (9.78, 0–40)	$t(84) = 0.57$, $p = 0.57$
Study Characteristics				
Implementation Strategy Condition (E-ATTC)	49 (57.0 %)	19 (67.9 %)	30 (51.7 %)	$\chi^2(1) = 2.00$, $p = 0.16$
First Role Play Fidelity Criterion CMCS Score	5.37 (0.75, 3.3 – 6.8)	5.48 (0.64, 4.3–6.8)	5.32 (0.81, 3.3–6.6)	$t(79) = -0.95$, $p = 0.34$

Notes.

[*] M (SD, Range) or n (%) respectively,

[1] Chi-Square Fisher's Exact Test was used to account for low cell frequency; CMCS = Contingency Management Competency Scale; E-ATTC = Enhanced - Addiction Technology Transfer Center

Table 2

Frequency CM fidelity according to provider and observer report.

CM Specific Practices To what extent did the provider...	Provider Report		Observer Report	Percent Agreement
	All Sessions (n = 3143 sessions, 86 providers)	Audio Recorded Sessions, (n = 72 sessions, 28 providers)	Audio Recorded Sessions (n = 72 sessions, 28 providers)	
Item 1) ...discuss how many prize draws were earned at this session?	3131 (99.6 %)	72 (100 %)	71 (98.6 %)	98.6 %
Item 2) ...discuss how many prize draws would be possible at the next CM session, if the patient meets the attendance target?	3111 (99.0 %)	72 (100 %)	70 (97.2 %)	97.2 %
Item 3) ...administer reinforcement?	3130 (99.6 %)	72 (100 %)	71 (98.6 %)	98.6 %
Item 4) ...assess patient's desire for prizes?	3067 (97.6 %)	71 (98.6 %)	61 (84.7 %)	83.3 %
Item 5) ...complement or praise the patient's efforts towards meeting the attendance target?	3118 (99.2 %)	72 (100 %)	65 (90.3 %)	90.3 %
Item 6) ...tie attendance and the CM program to abstinence and other treatment goals?	3084 (98.1 %)	70 (97.2 %)	42 (58.3 %)	61.1 %
All six items present in session	2998 (95.4 %)	69 (95.8 %)	35 (48.6 %)	34 (47.2 %)

Notes. CM = contingency management